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Medicinal plants with antimalarial properties found in Tonto Block west singhbhum Jharkhand India

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Abstract- Malaria is a mosquito-borne infectious disease of humans and other animals caused by protists (a type of microorganism) of the genus *Plasmodium*. According to the guidelines recommended by the National Vector Borne Disease Control Programme, phytochemical-based therapy is given priority as the first line of treatment for malaria in rural areas. Tribal people living in rural areas use medicinal plants as drugs to reduce the symptoms of malaria, which remains a major health problem. More than 5% of the population was found to be infected with malaria during random surveys. The percentage increases during the summer season due to resistance to antimalarial injectable and oral drugs, as well as the presence of multiple species of vectors and protozoans. To fight existing problems and drug resistance, local traditional healers (hakims) use plant extracts and administer doses according to the body weight of patients. Plant extracts are used for the treatment of malarial fever, loss of appetite, diabetes, joint pain, jaundice, digestive tract infections, and skin-related diseases. The present study focuses on identifying plants found in the Tonto Block of West Singhbhum, Jharkhand, that possess antimalarial, medicinal, and bioactive properties and are used by tribal and ethnic communities living in the area. Many plants with antimalarial properties have been identified, which have been used by tribal people for centuries and are still in use. These plants are bioactive and contain compounds such as terpenoids, tannins, alkaloids, flavonoids, anthraquinones, and pyrroles. The study mainly focuses on gathering information about bioactive plants used as antimalarial remedies.

Keywords: Antimalarial plants, antiplasmodial activity, tribal medicine, ethnobotany, bioactive compounds, traditional healthcare, malaria control

INTRODUCTION

Malaria is a major parasitic disease and a serious public health problem caused by parasites known as *Plasmodium vivax, Plasmodium falciparum, Plasmodium malariae,* and *Plasmodium ovale* in the Tonto Block, West Singhbhum area, which has a variety of medicinal plants. Man develops malaria after 10 to 14 days of being bitten by an infected mosquito. *Plasmodium falciparum* is the main protozoan reported to cause malaria.

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Medicinal plants are not only used in treating malaria but also for the treatment of cattle and as biological pest insecticides.

Tonto is one of the blocks of West Singhbhum district, Jharkhand, contributing a population of about 83,379, most of whom are tribes living in remote areas.

Forest cover in Jharkhand is 29.62% of the total geographical area (Anonymous, n.d.), and in the Tonto Block, forest cover is maximal, including shrubs and herbs, making the area highly humid and water-containing, which provides good conditions for malarial vectors to breed.

There are more than 18,000 flowering plants, of which more than 8,000 species are medicinally important. The block is populated by Scheduled Tribes and Scheduled Castes, including Ho, Santhal, Bhumij, Munda, Orao, etc. People are largely dependent on forests to feed, live, and cure.

Medicinal plants are bioactive and contain secondary metabolites used to cure malarial fever in humans and are largely used to cure various types of diseases.

Change in the climatic condition of the Tonto Block is such that it favors the growth of malarial vectors, eventually leading to a rise in the annual parasite index, which is calculated with respect to the positive slides for the year and the population of that area. Warm and humid conditions in the Tonto region support the growth of malarial vectors.

Currently, 80.5% of the 1.09 billion population of India lives in malaria-risk areas. Of this, 4.2%, 32.5%, and 43.8% live in areas of high, moderate, and low risk to malaria, respectively.[1]

The history of the disease shows that the Global Malaria Eradication Programme of WHO, launched in the 1950s, was a huge success in India, as the incidence declined from an estimated 75 million cases and 800,000 deaths in 1947 to just 49,151 cases (annual parasite incidence per thousand [API]: 0.13; slide positivity rate [SPR]: 0.38%; and *Plasmodium falciparum* [*Pf*]: 34.9%) and no deaths in 1961, and malaria was thought to be on the verge of eradication. It was then that a series of setbacks were witnessed, leading to malaria resurgence in multiple foci in the country, and reported cases increased to 1,322,398 by 1971 (API: 2.47; SPR: 3.27%; and Pf: 11.2%) and then to 6,467,215 in 1976 (API: 11.25; SPR: 11.6%; and Pf: 11.7%).[1]

In West Singhbhum, for the year 2024, the total blood slide collection was 294,636, of which 16,617 slides were positive, with the district API at 8.8, Pf% at 90.50%, and SPR at 4.02%.

It begins with a bite from an infected female mosquito, which introduces the parasites via its saliva into the circulatory system and ultimately to the liver, where they mature and reproduce. The disease causes symptoms that typically include fever and headache, which in severe cases can progress to coma or death. The term malaria originates from Medieval Italian mala aria, meaning 'bad air'; the disease was formerly called ague or marsh fever due to its association with swamps and marshlands.

The burden due to *Plasmodium falciparum* malaria has reduced over the years with a decline in malaria cases following the use of ACT after reports of multidrug resistance in the area. Some high-risk areas are covered with LLINs and IRS, which gave good results in the past few years by reducing the API. The use of RDTs in remote areas has also helped in early diagnosis and complete treatment to control malarial disease. *Plasmodium falciparum* infection contributes most to severe malaria, which, if not treated, may lead to death.

The block has hills alternating with valleys, steep mountains, and forests on mountain slopes. Rainfall is highest in July and August, and the annual rainfall in the block is about 1,422 mm. April and May are the hottest months, while December and January are the coldest months.

Table 1- Antimalarial medicinal plants of Tonto Block West Singhbhum

Sl. No	Plant Species	Common Name	Part used	Traditional Uses
1	*Andrographis paniculata* 	King of Bitters, Kalmegh	Whole plant	Antimalarial, antipyretic, anti-inflammatory, antidiabetic, antihypertensive, antihepatotoxic.[2] Andrographolide ($C_{20}H_{30}O_5$), an active component of *Andrographis paniculata* (Burm.f.) Wall. ex Nees, has been used for centuries in traditional medical systems throughout Asia. Andrographolide's antimalarial characteristics were reported in the late twentieth century. *Andrographis* has traditionally been used to treat fevers associated with malaria. Initial studies focused on in vitro activity against malaria parasites. Later studies confirmed inhibitory effects against *Plasmodium falciparum*. Animal studies also supported its antimalarial potential

2	*Azadirachta indica* 	Neem	Leaf, Stem Bark, Roots, Fruits	Fever, malaria, headache, ulcers, diabetes, leprosy, chickenpox, skin diseases.[3] Leaf extract has been prescribed orally for malaria treatment in Ayurveda. It contains sterols, limonoids, flavonoids, glycosides, and coumarins. The extract shows schizontocidal and gametocidal effects. Combination with artesunic acid may reduce malaria transmission. Efficacy was evaluated against *Plasmodium berghei* using Swiss albino mice. Azadirachtin, a phytochemical isolated from *A. indica*, is used in the treatment of cerebral malaria.
3	*Cassia alata* 	Ringworm Shrub, Senna,	Leaf, Seeds, Stem, Flower	Eczema, ringworm, malaria, constipation. Leaf extract is used as an antimalarial due to the presence of anthraquinones, alkaloids, tannins, flavonoids, terpenoids, and steroids. The antimalarial mechanism is not yet fully understood. In vitro activity was evaluated against *Plasmodium falciparum* (strain D10), and in vivo activity was assessed using *Plasmodium berghei* in mice. Dichloromethane/methanol (1:1) extract showed significant reduction in parasitemia
4	*Berberis aristata* 	Barberry, Daru Haldi, Daruharidra	Root	Malaria, skin diseases, diarrhea, eye problems, inflammation, wound healing. The plant contains berberine, which exhibits antimalarial properties and is used in malaria treatment.[4]
5	*Cissampelos pareira* 	Velvetleaf, Harjori or Akanadi	Root, Stem	Malaria, ulcer, wound, rheumatism, asthma, cholera, diarrhoea, snakebite, rabies. Benzylisoquinoline and bisbenzylisoquinoline alkaloids are the major phytochemicals. Leaf juice enriched with flavonoids has been traditionally used to treat malarial fever. In silico molecular docking studies showed good binding affinity of flavonoids with antimalarial targets.

6	*Cassia occidentalis* 	Coffee Senna, Chakod	Root bark	Fever, laxative, whooping cough, skin diseases, asthma, joint pain. Contains achrosin, aloe-emodin, emodin, anthraquinones, anthrones, apigenin, aurantiobtusin, campesterol, cassiollin, chrysophanol, chrysoeriol, physicon, quercetin, rhamnosides, rhein, sitosterols, tannins, and xanthorine. The plant exhibits anti-inflammatory, antihepatotoxic, antibacterial, and antiplasmodial activities.
7	*Curcuma longa* 	Turmeric, Haldi	Rhizome	Malaria, cancer, skin diseases, joint pain. Ethanolic rhizome extract showed *in vitro* antimalarial activity. The isolated active compound exhibited schizont suppression of 62.63% at 0.625 µg/mL with IC_{50} values below 0.625 µg/mL.
8	*Holarrhena pubescens* 	Fever pod, Indrajau, kurchi	Root, Bark, Seed	Malarial fever, diabetes, piles, leprosy, diarrhea. Exhibits antimalarial activity against *Plasmodium falciparum*. Contains alkaloids, tannins, flavonoids, and glycosides contributing to therapeutic effects.
9	*Datura metel* 	Thornapple, Datura	Fruit	Malaria, dandruff, arthritis, liver disorders, viral fever, antifungal. Methanolic extract showed significant inhibition of parasitemia in *Plasmodium berghei*-infected mice. Antivector activity was also reported against *Anopheles stephensi*.

10	*Lantana camara* 	Lantana, Putoos	Leaf	Fever, cold, rheumatism, asthma, cuts, intestinal worms.[5] Leaf extract demonstrated in vivo antimalarial activity.
11	*Moringa oleifera* 	Drumstick Tree, Sahjan	Leaf, Bark, Flower, Fruit, Seed	Malaria, malnutrition, water purification, joint pain, diabetes, hypertension, wound healing.[6] Flavonoids such as quercetin, apigenin, and rutin extracted from leaves and seeds show antimalarial activity.
12	*Eclipta alba* 	False Daisy, Bhringaraj	Leaf	Malaria, fever, hair loss, jaundice, skin diseases.[7] Methanolic leaf extract showed dose-dependent chemosuppression against *Plasmodium berghei* in mice with increased mean survival time.
13	*Ocimum sanctum* 	Holy Basil, Tulsi	Bark, Leaves, Flowers, Roots, Fruits, Seeds	Malarial fever, cold, headache, stomach disorders, inflammation, heart disease. Eugenol contributes to antimalarial and repellent properties.

14	*Solanum incanum* 	Bitter Apple, Jangli Bhata, Sodom apple	Leaf, Fruit, Root	Malaria, sore throat, stomach ache, headache, painful menstruation, diabetes. Leaf extract down-regulated δ-aminolevulinate dehydratase in *Plasmodium berghei*-infected mice.
15	*Plumbago zeylanica* 	Ceylon leadwort, Chitrak	Root, Aerial Part	Anti-plasmodial, anti-tumour, antibacterial, anti-inflammatory. Bioactive compounds include plumbagin, sitosterol, elliptinone, and zeylanone.
16	*Pongamia pinnata* 	Indian Beech, Karanj	Leaf, stem, bark, flower, fruit, kernel, latex	Malaria, fever, wounds, skin diseases, diabetes. Rich in flavonoids, alkaloids, glycosides, and fixed oils.
17	*Aegle marmelos* 	Wood Apple, Bael	Leaf, Seed	Antipyretic, antidiabetic, analgesic, anti-inflammatory. Leaf methanol extract showed activity against *Plasmodium falciparum*.

18	*Artemisia indica* 	Wormwood, Majtari	Whole plant	Malarial fever, tuberculosis, wounds, dysentery. Artemisinin from *Artemisia annua* is used as artemisinin combination therapy (ACT) for malaria treatment

CONCLUSION

As malaria parasites continue to develop resistance to many commonly used antimalarial drugs, there is increasing interest in the use of plant-based remedies for the treatment and management of malaria. Bioactive plant species play an important role in addressing the problem of drug resistance and offer promising alternative sources of antimalarial compounds. In many rural areas, dried plant materials are traditionally used as mosquito repellents, particularly during the evening hours. The West Singhbhum region is rich in plant species containing bioactive constituents, and further systematic exploration is required to identify and evaluate their potential antimalarial properties.

FUTURE PROSPECTIVES

The search for antimalarial plants and their bioactive compounds, along with focused research and development of novel drugs, can help address the life-threatening burden of malaria in developing countries, including India.

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